Timothy Agboada

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Education

PhD Computational Data Science and Engineering — Aug 2025 – May 2028
North Carolina A&T University, Greensboro, NC

Master of Science in Computer Science — Jan 2022 – Dec 2024
Norfolk State University, Norfolk, VA

Bachelor of Science in Computer Science and Engineering — Sept 2014 – May 2018
University of Mines and Technology, Tarkwa, Ghana

Skills

AI & ML Engineering: LLM Orchestration (LangChain, LangGraph), RAG Pipelines, Prompt Engineering, Agentic Workflows, PyTorch, Scikit-learn, Mathematical Modeling.

Languages & Tools: Python, C++, MATLAB, SQL, Java, Docker, Kubernetes, Git, AWS (S3, Lambda, Redshift).

Core Competencies: Algorithm Optimization, Scalable System Design, Statistical Inference, Geospatial Analysis (ArcGIS, Pix4Dmapper), Parallel Computing (OpenMP).

Work Experience

Applied Science Researcher — Mar 2025 – Aug 2025
Weber State University — *Ogden, UT*

- Developed and evaluated novel base models for parallel tropical computations, improving server computational performance and algorithmic efficiency by 19%.

Software Development Engineer — Aug 2024 – Dec 2024
Amazon Robotics — *Westborough, MA*

- Optimized robotic workflows by applying advanced simulation modeling and analyzing complex system performance metrics, improving operational efficiency by 15%.
- Engineered and executed comprehensive system testing protocols to ensure 100% data reliability and robustness across automated environments.

Data Engineer — June 2023 – Aug 2023
Singlestone Consulting — *Richmond, VA*

- Automated data ingestion and staging frameworks using Python, DBT, and AWS Glue, accelerating processing speed by 40%.
- Built robust data validation frameworks for machine learning pipelines, reducing systemic data inconsistencies by 30%.

Customer and Marketing Data Analyst — Sep 2019 – Dec 2021
British Council — *Accra, Ghana*

- Developed statistical segmentation and predictive behavioral models using Python and SQL, resulting in a 15% improvement in conversion rates.
- Designed and implemented large-scale A/B testing frameworks to evaluate and optimize marketing campaign performance.

Data Engineer — May 2018 – Sep 2019
Vodafone — *Accra, Ghana*

- Designed scalable data infrastructure and ETL processes utilizing Python, SQL, and Apache Spark, yielding a 30% improvement in system latency.
- Engineered features and preprocessed extensive datasets for predictive machine learning models, boosting downstream inference accuracy by 15%.

Projects

Autonomous EB-2 NIW Immigration Evaluator — Github — *Python, LangGraph, Gemini API, Streamlit*
Engineered a stateful, multi-node AI agent utilizing LangGraph to parse user academic profiles against complex Matter of Dhanasar legal criteria, outputting synthesized strategic plans and identifying evidence gaps via a dynamic UI.

Retrieval-Augmented Travel Planner Agent — Github — *Python, LangChain, DuckDuckGo Search, FPDF2*
Built a RAG pipeline integrating real-time web search with an LLM to autonomously orchestrate tool-use, reasoning over hard constraints (budget, dates) to generate optimized, exportable itineraries.

UAV Flood Mapping & Visual Question Answering — *Python, PyTorch, Deep Learning*
Developed deep learning VQA architectures applied to UAV remote sensing imagery using DeepFlood and InfraFlood-NC datasets.

Geospatial Elevation Validation & Scalable Processing — *C++, OpenMP, ArcGIS Pro*
Evaluated UAV-derived DEMs against LiDAR ground truth and implemented parallelized heap sort algorithms for scalable data processing.

Leadership, Awards, & Publications

Awards: NSU Outstanding Graduate Student Award (April 2024); ASPIRE Mentor (2023–2024).

Certifications: AWS Certified Data Engineer, AWS Cloud Practitioner, Professional Scrum Master I.

Publications: Oral Presenter, IGARSS 2026 Conference: Visual Question Answering in Remote Sensing; Presenter, ICCA 2024 Conference [IEEE publication].